

Friday, February 14

Solutions for Heteroscedasticity

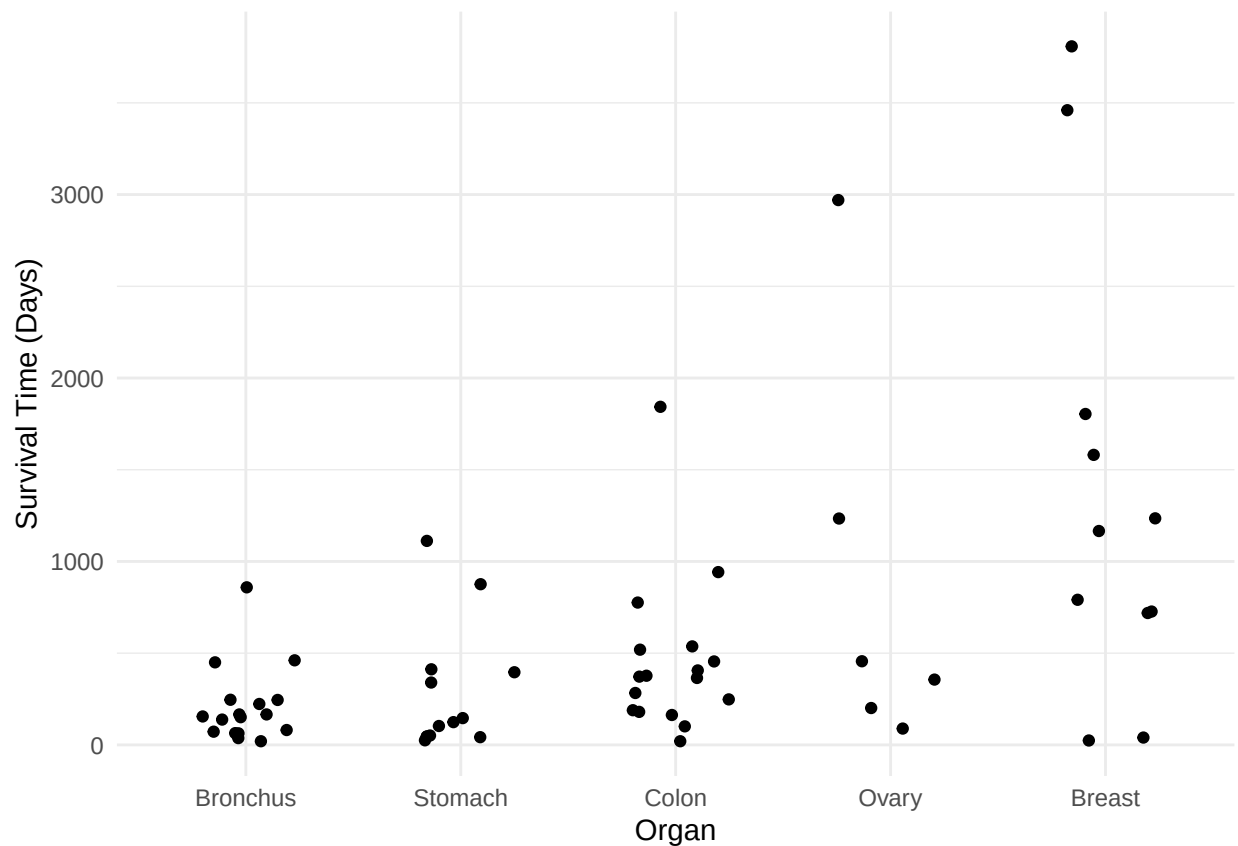
We will discuss four solutions to heteroscedasticity in linear and nonlinear regression: variance-stabilizing transformations, weighted least squares, robust standard errors, and models that do not assume homoscedasticity.

Variance-Stabilizing Transformations

The idea is to use $Y_i^* = g(Y_i)$ instead of Y_i as the response variable, where g is a *variance-stabilizing transformation*.

Example: Consider again the cancer survival time data.

```
library(Stat2Data)
data(CancerSurvival)
CancerSurvival$Organ <- with(CancerSurvival, reorder(Organ, Survival, mean))
p <- ggplot(CancerSurvival, aes(x = Organ, y = Survival)) +
  geom_jitter(height = 0, width = 0.25) +
  labs(y = "Survival Time (Days)") + theme_minimal()
plot(p)
```



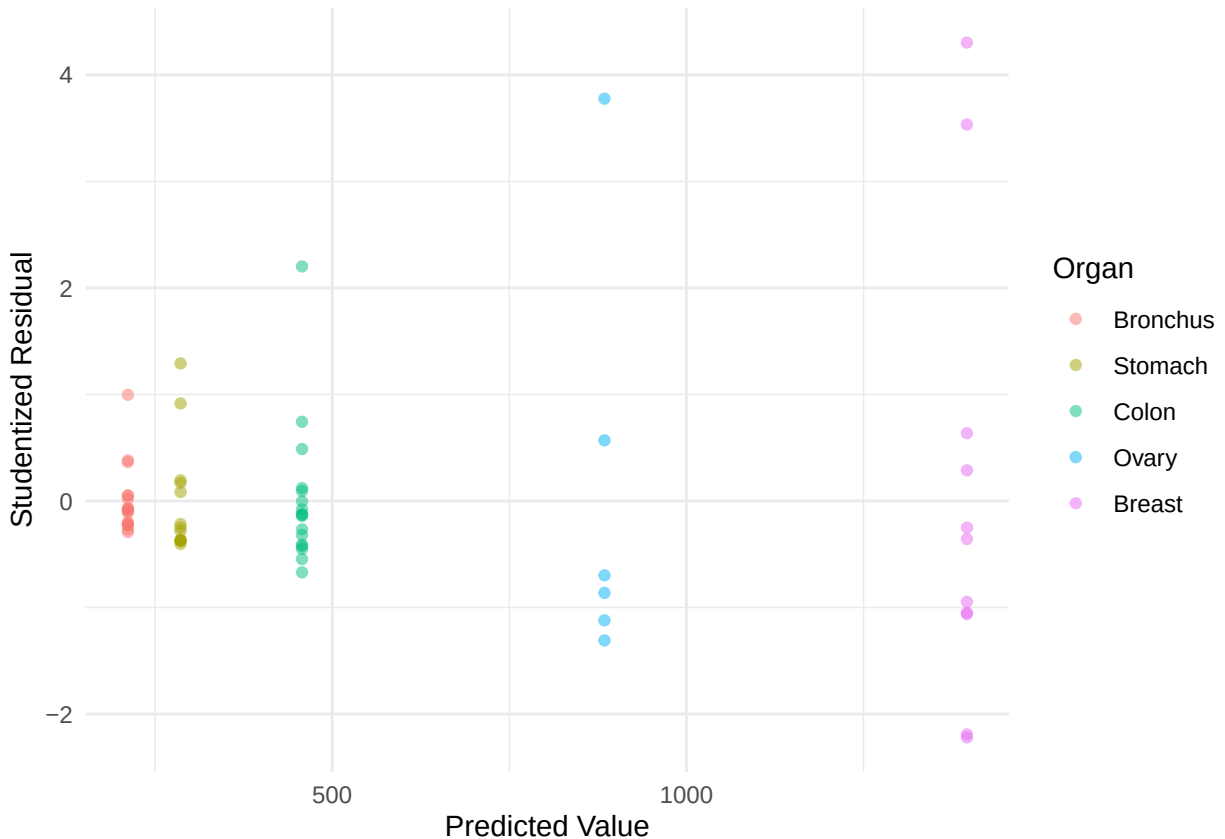
```

m <- lm(Survival ~ Organ, data = CancerSurvival)

CancerSurvival$yhat <- predict(m)
CancerSurvival$rest <- rstudent(m)

p <- ggplot(CancerSurvival, aes(x = yhat, y = rest, color = Organ)) +
  geom_point(alpha = 0.5) + theme_minimal() +
  labs(x = "Predicted Value", y = "Studentized Residual")
plot(p)

```

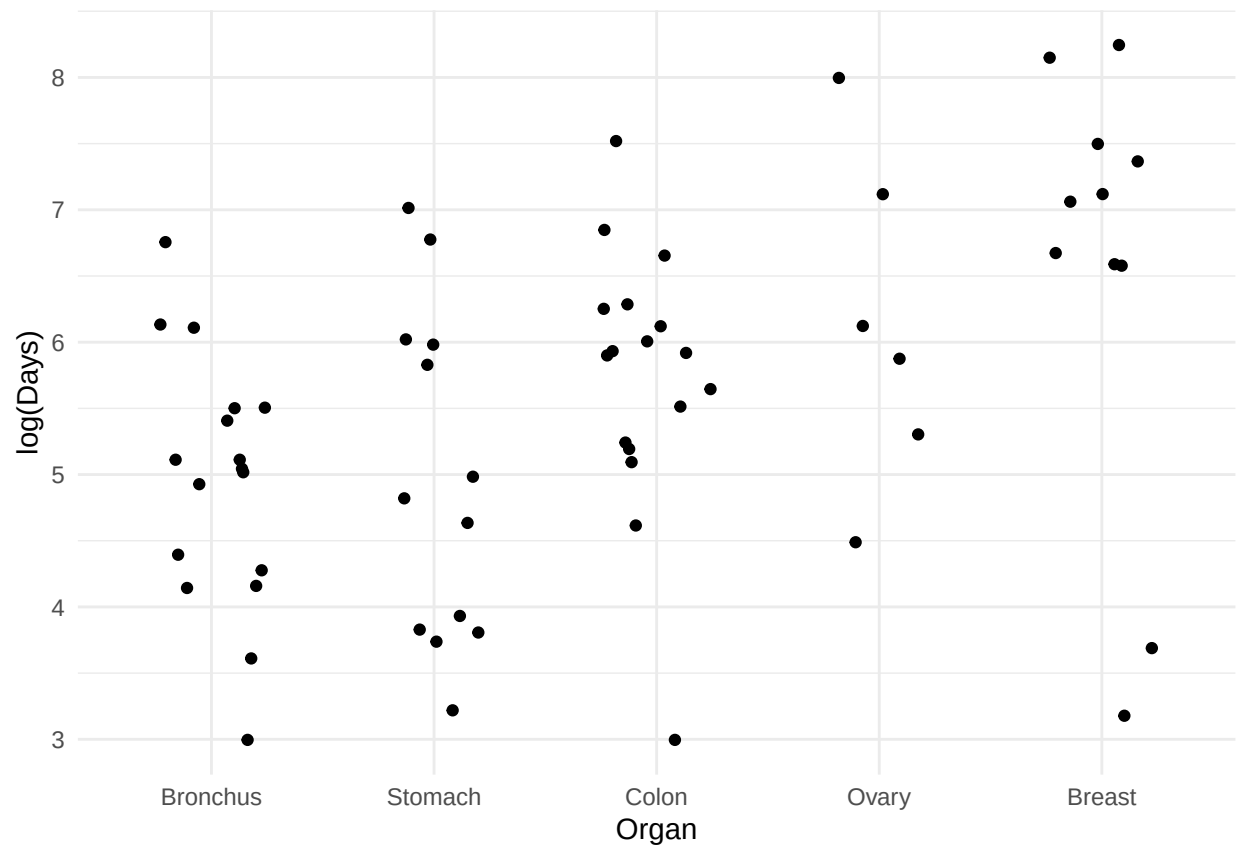


A model for *log* time might exhibit something closer to homoscedasticity.

```

p <- ggplot(CancerSurvival, aes(x = Organ, y = log(Survival))) +
  geom_jitter(height = 0, width = 0.25) +
  labs(y = "log(Days)") + theme_minimal()
plot(p)

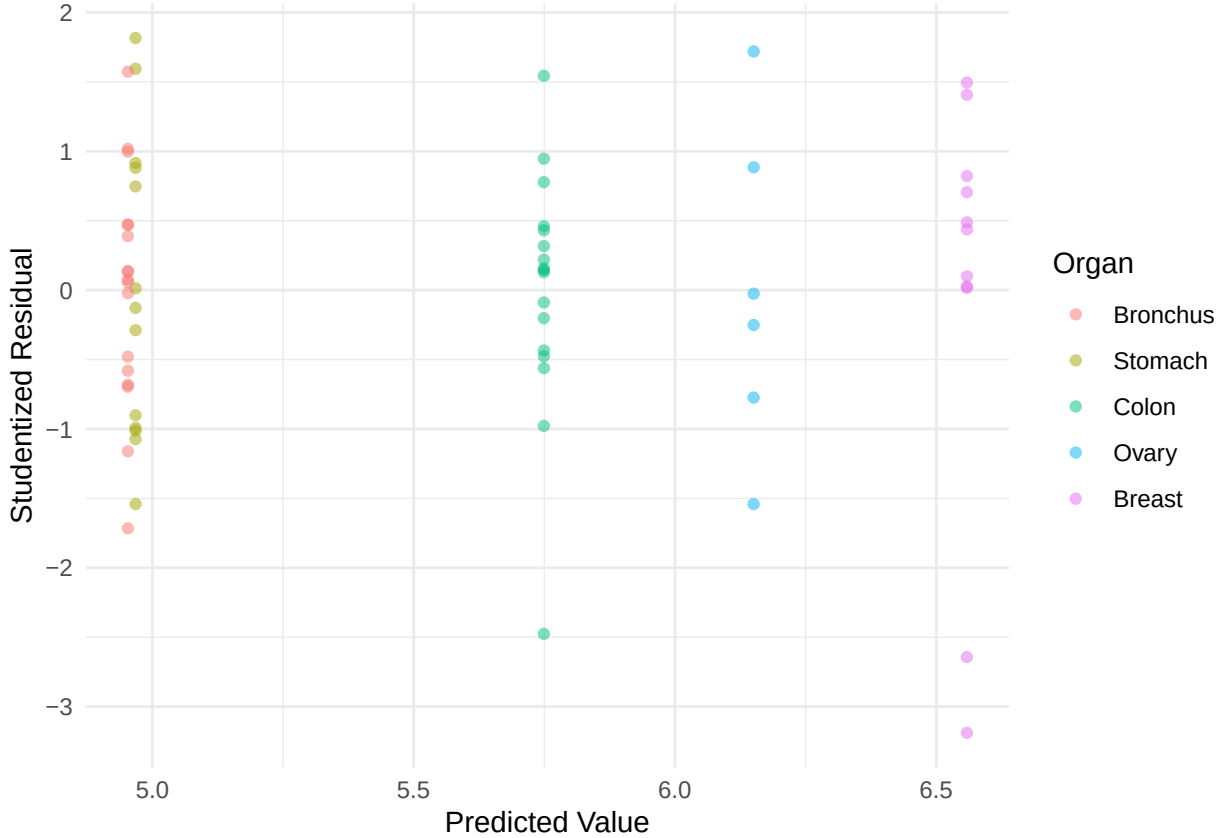
```



```
m <- lm(log(Survival) ~ Organ, data = CancerSurvival)

CancerSurvival$yhat <- predict(m)
CancerSurvival$rest <- rstudent(m)

p <- ggplot(CancerSurvival, aes(x = yhat, y = rest, color = Organ)) +
  geom_point(alpha = 0.5) + theme_minimal() +
  labs(x = "Predicted Value", y = "Studentized Residual")
plot(p)
```



Comments on variance-stabilizing transformations.

1. Depending on the situation, other transformations may exhibit variance-stabilizing properties. Some common transformations are $\sqrt{Y_i}$, $\log Y_i$, $1/\sqrt{Y_i}$ and $1/Y_i$ for right-skewed response variables, and $n_i \arcsin(\sqrt{Y_i})$ when Y_i is a proportion with a denominator of n_i .
2. A limitation of variance stabilizing transformations is that it is often difficult (and undesirable) to *interpret* the model in terms of the transformed response variable (although there are exceptions as we will later see with the log transformation in the context of accelerated failure time models for survival data).
3. It is important to note that for any *nonlinear* transformation that $E[g(Y)] \neq g[E(Y)]$ (i.e., the expected transformed response does not necessarily equal the transformed expected response). For example, $E[\log Y] \neq \log E(Y)$. So we cannot obtain inferences for the expected response by applying the inverse of the transformation function. For example, $\exp[E(\log Y)] \neq E(Y)$.

Weighted Least Squares

A *weighted* least squares (WLS) estimator of the regression model parameters minimizes

$$\sum_{i=1}^n w_i (y_i - \hat{y}_i)^2,$$

where $w_i > 0$ is the *weight* for the i -th observation. So-called *ordinary least squares* (OLS) or *unweighted least squares* is a special case where all $w_i = 1$.

To account for heteroscedasticity, the weights should be *inversely proportional to the variance of the response* so that

$$w_i \propto \frac{1}{\text{Var}(Y_i)}.$$

Estimation is *efficient* meaning that the *true* standard errors (which are not necessarily the *reported* standard errors shown by software since these are estimates and may be biased without using weights as defined above) are as small as they can be when using weighted least squares.

Example: Consider the following data.

```
turkeys <- data.frame(
  weight = c(674, 764, 795, 796, 826, 782, 834, 836, 830),
  pens = c(10, 5, 2, 2, 5, 5, 2, 2, 5),
  dosea = c(c(0, 0.12, 0.22, 0.32, 0.44), rep(0, 4)),
  doseb = c(rep(0, 5), c(0.12, 0.22, 0.32, 0.44))
)
turkeys
```

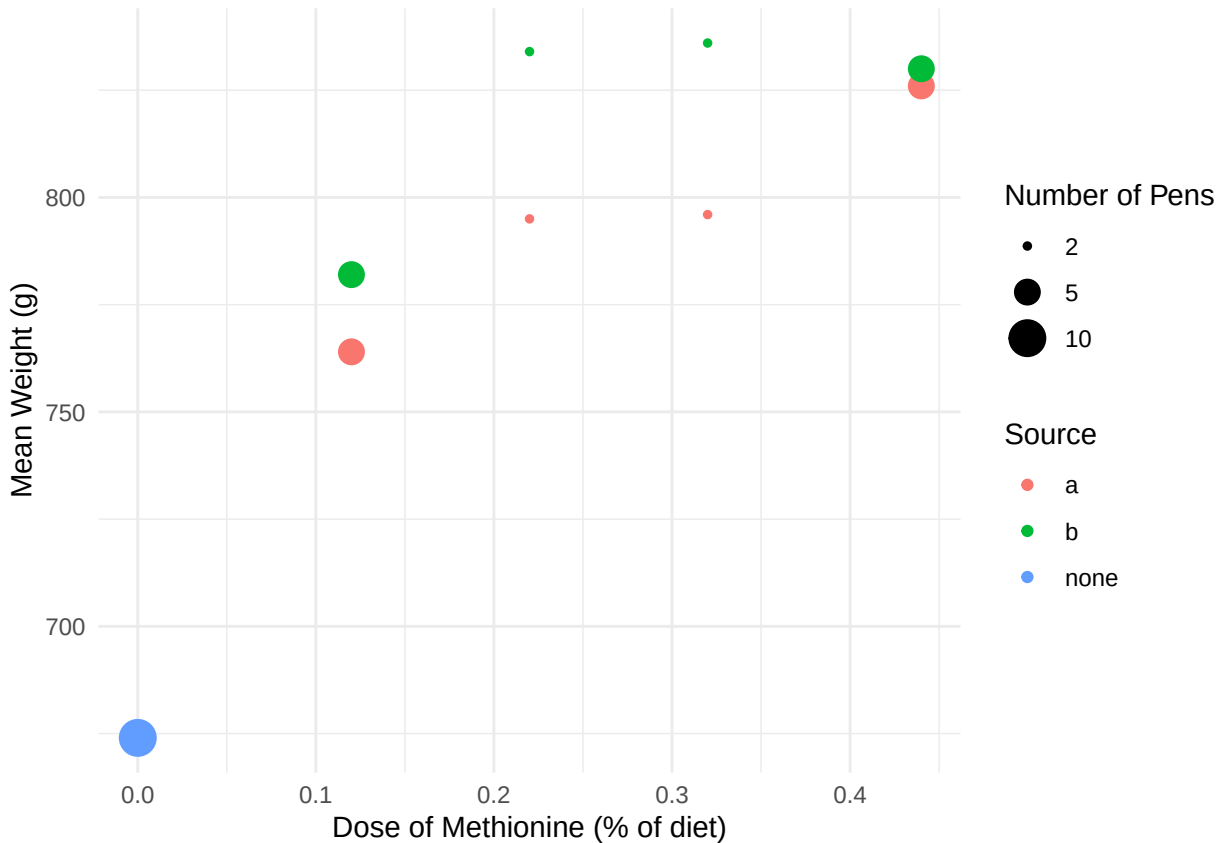
	weight	pens	dosea	doseb
1	674	10	0.00	0.00
2	764	5	0.12	0.00
3	795	2	0.22	0.00
4	796	2	0.32	0.00
5	826	5	0.44	0.00
6	782	5	0.00	0.12
7	834	2	0.00	0.22
8	836	2	0.00	0.32
9	830	5	0.00	0.44

For plotting and modeling convenience we will rearrange the data a bit.

```
library(dplyr)
turkeys <- turkeys |>
  mutate(dose = dosea + doseb) |>
  mutate(source = case_when(
    dose == 0 ~ "none",
    dosea > 0 ~ "a",
    doseb > 0 ~ "b")
  ) |>
  select(-dosea, -doseb)
turkeys
```

	weight	pens	dose	source
1	674	10	0.00	none
2	764	5	0.12	a
3	795	2	0.22	a
4	796	2	0.32	a
5	826	5	0.44	a
6	782	5	0.12	b
7	834	2	0.22	b
8	836	2	0.32	b
9	830	5	0.44	b

```
library(ggplot2)
p <- ggplot(turkeys, aes(x = dose, y = weight, color = source)) +
  theme_minimal() + geom_point(aes(size = pens)) +
  scale_size(breaks = c(2, 5, 10)) +
  labs(x = "Dose of Methionine (% of diet)", y = "Mean Weight (g)",
    color = "Source", size = "Number of Pens")
plot(p)
```



Suppose we want to estimate the following model.

$$E(W_i) = \begin{cases} \gamma, & \text{if no methionine was given,} \\ \alpha + (\gamma - \alpha)2^{-d_i/\lambda_a}, & \text{if methionine was given from source } a, \\ \alpha + (\gamma - \alpha)2^{-d_i/\lambda_b}, & \text{if methionine was given from source } b. \end{cases}$$

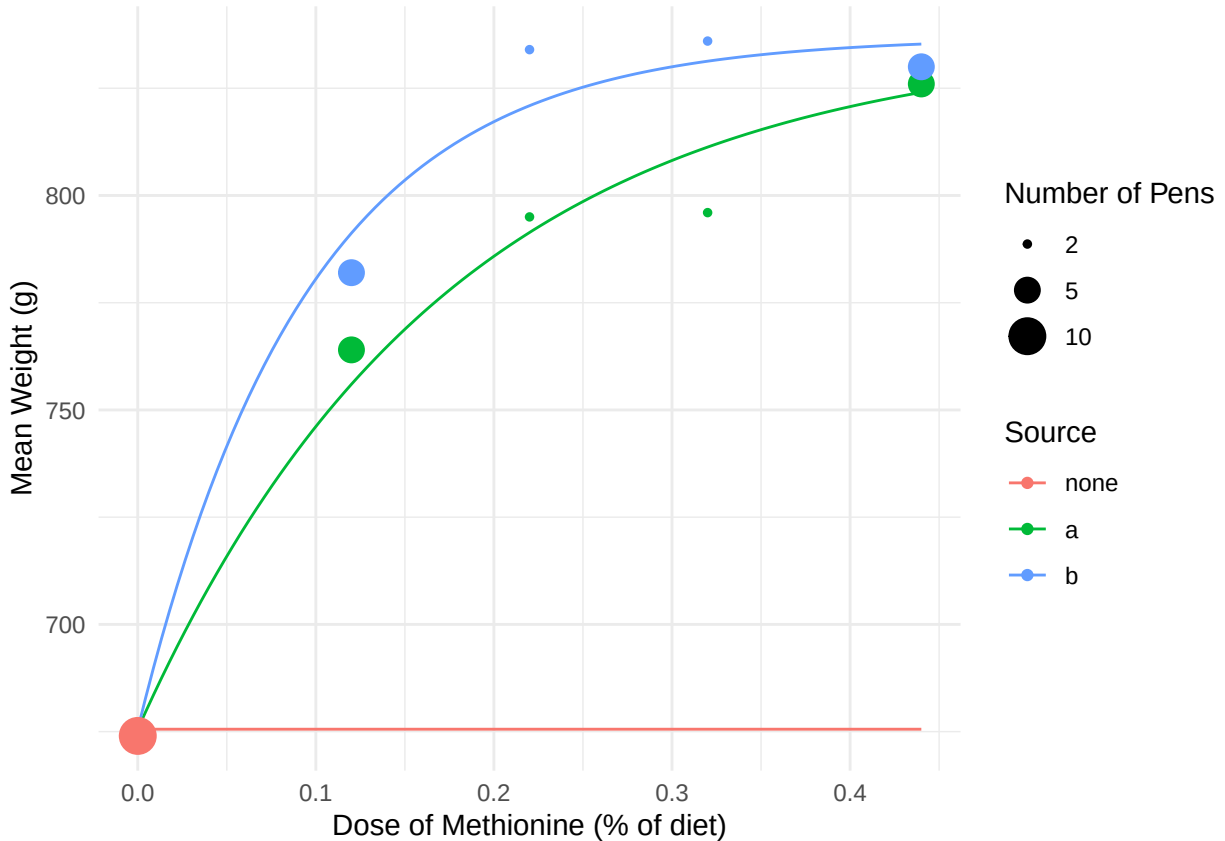
Note that this is like the Von Bertalanffy growth model. Here α is an asymptote for expected weight that we approach as dose increases, γ is the expected response for a zero dose, and λ_a and λ_b are “half-life” parameters. We can estimate this model as follows using *ordinary* (i.e., unweighted) least squares as follows.

```
m.ols <- nls(weight ~ case_when(
  source == "none" ~ gamma,
  source == "a" ~ alpha + (gamma - alpha) * 2^(-dose/lambdaa),
  source == "b" ~ alpha + (gamma - alpha) * 2^(-dose/lambdab)),
  start = list(alpha = 825, gamma = 675, lambdaa = 0.1, lambdab = 0.1),
  data = turkeys)
summary(m.ols)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
alpha	836.87362	8.87663	94.278	2.545e-09
gamma	675.56291	10.92399	61.842	2.092e-08
lambdaa	0.12049	0.02250	5.355	3.051e-03
lambdab	0.06589	0.01601	4.115	9.222e-03

```
d <- expand.grid(source = c("none", "a", "b"), dose = seq(0, 0.44, length = 100))
d$yhat <- predict(m.ols, newdata = d)
p <- ggplot(turkeys, aes(x = dose, y = weight, color = source)) +
  geom_line(aes(y = yhat), data = d) +
```

```
theme_minimal() + geom_point(aes(size = pens)) +
scale_size(breaks = c(2, 5, 10)) +
labs(x = "Dose of Methionine (% of diet)", y = "Mean Weight (g)",
color = "Source", size = "Number of Pens")
plot(p)
```



The response variable is an *mean* of several observations so that

$$Y_i = \frac{Z_{i1} + Z_{i2} + \cdots + Z_{in_i}}{n_i}$$

where Z_{ij} is the length of the j -th pen that goes into the i -th average, and a total of n_i pens go into the i -th average. If $\text{Var}(Z_{ij}) = \sigma^2$ then $\text{Var}(Y_i) = \sigma^2/n_i$. Thus the weights should be

$$w_i \propto \frac{1}{\sigma^2/n_i} = \frac{n_i}{\sigma^2}.$$

Assuming $1/\sigma^2$ is a constant for all observations, we can define the weights of the means as $w_i = n_i$. The weights can be specified in `lm` and `nls` (and other functions for regression) using the `weights` argument.

```
m.wls <- nls(weight ~ case_when(
  source == "none" ~ gamma,
  source == "a" ~ alpha + (gamma - alpha) * 2^(-dose/lambdaa),
  source == "b" ~ alpha + (gamma - alpha) * 2^(-dose/lambdab)),
start = list(alpha = 825, gamma = 675, lambdaa = 0.1, lambdab = 0.1),
data = turkeys, weights = pens)
summary(m.wls)$coefficients
```

Estimate Std. Error t value Pr(>|t|)

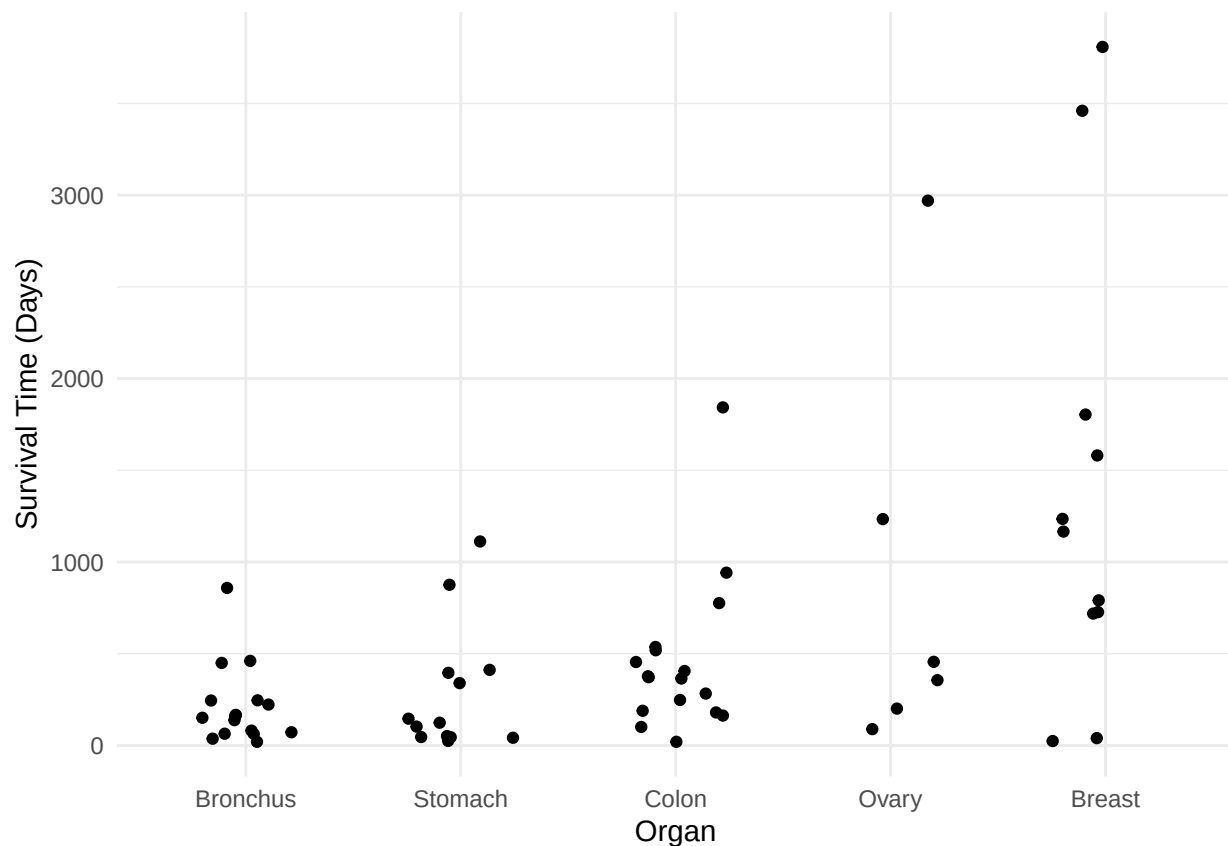
```
alpha 834.78331      6.7761 123.195 6.684e-10
gamma 674.30393      5.5420 121.671 7.113e-10
lambdaa 0.10997      0.0163 6.746 1.086e-03
lambdab 0.06857      0.0117 5.860 2.051e-03
```

```
summary(m.ols)$coefficients
```

```
      Estimate Std. Error t value Pr(>|t|)
alpha 836.87362     8.87663  94.278 2.545e-09
gamma 675.56291    10.92399  61.842 2.092e-08
lambdaa 0.12049     0.02250   5.355 3.051e-03
lambdab 0.06589     0.01601   4.115 9.222e-03
```

Example: Consider again the cancer survival time data.

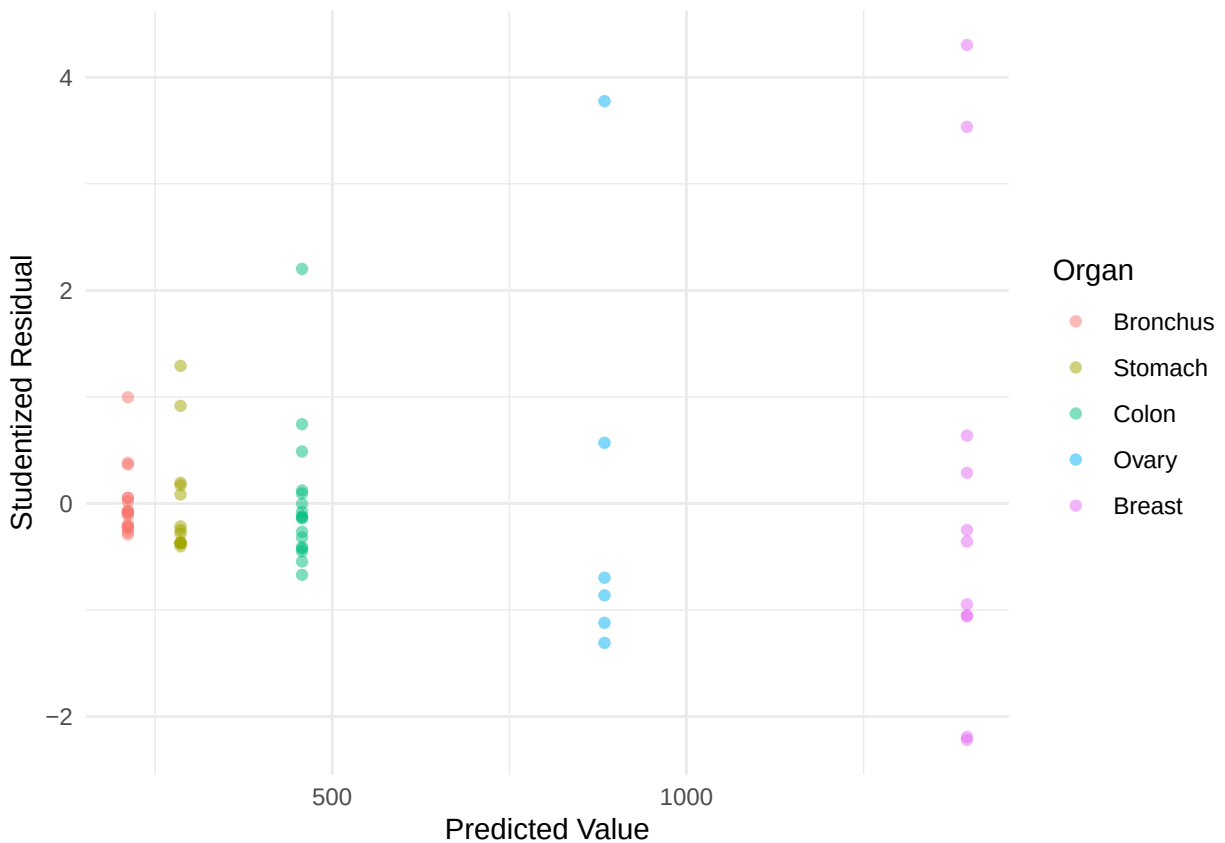
```
p <- ggplot(CancerSurvival, aes(x = Organ, y = Survival)) +
  geom_jitter(height = 0, width = 0.25) +
  labs(y = "Survival Time (Days)") + theme_minimal()
plot(p)
```



```
m.ols <- lm(Survival ~ Organ, data = CancerSurvival)

CancerSurvival$yhat <- predict(m.ols)
CancerSurvival$rest <- rstudent(m.ols)

p <- ggplot(CancerSurvival, aes(x = yhat, y = rest, color = Organ)) +
  geom_point(alpha = 0.5) + theme_minimal() +
  labs(x = "Predicted Value", y = "Studentized Residual")
plot(p)
```

There are a couple of ways we could go with these data. One is that since we have a categorical explanatory variable with multiple observations per category, we could *estimate* the variance of Y_i of each organ, and then set the weights to the reciprocals of these estimated variances.

```
library(dplyr)
CancerSurvival |> group_by(Organ) |>
  summarize(variance = var(Survival), weight = 1/var(Survival))
```

```
# A tibble: 5 x 3
  Organ    variance    weight
  <fct>    <dbl>    <dbl>
1 Bronchus  44041. 0.0000227
2 Stomach   119930. 0.00000834
3 Colon    182473. 0.00000548
4 Ovary    1206875. 0.000000829
5 Breast   1535038. 0.000000651
```

We can use the following to compute weights and add them to the data frame.

```
CancerSurvival <- CancerSurvival |>
  group_by(Organ) |> mutate(w = 1/var(Survival))
head(CancerSurvival)
```

```
# A tibble: 6 x 3
# Groups:   Organ [1]
  Survival Organ    w
  <int> <fct>    <dbl>
1    124 Stomach 0.00000834
2     42 Stomach 0.00000834
```

```

3      25 Stomach 0.00000834
4      45 Stomach 0.00000834
5     412 Stomach 0.00000834
6      51 Stomach 0.00000834

```

Now let's estimate the model using weighted least squares with these weights and inspect the residuals.

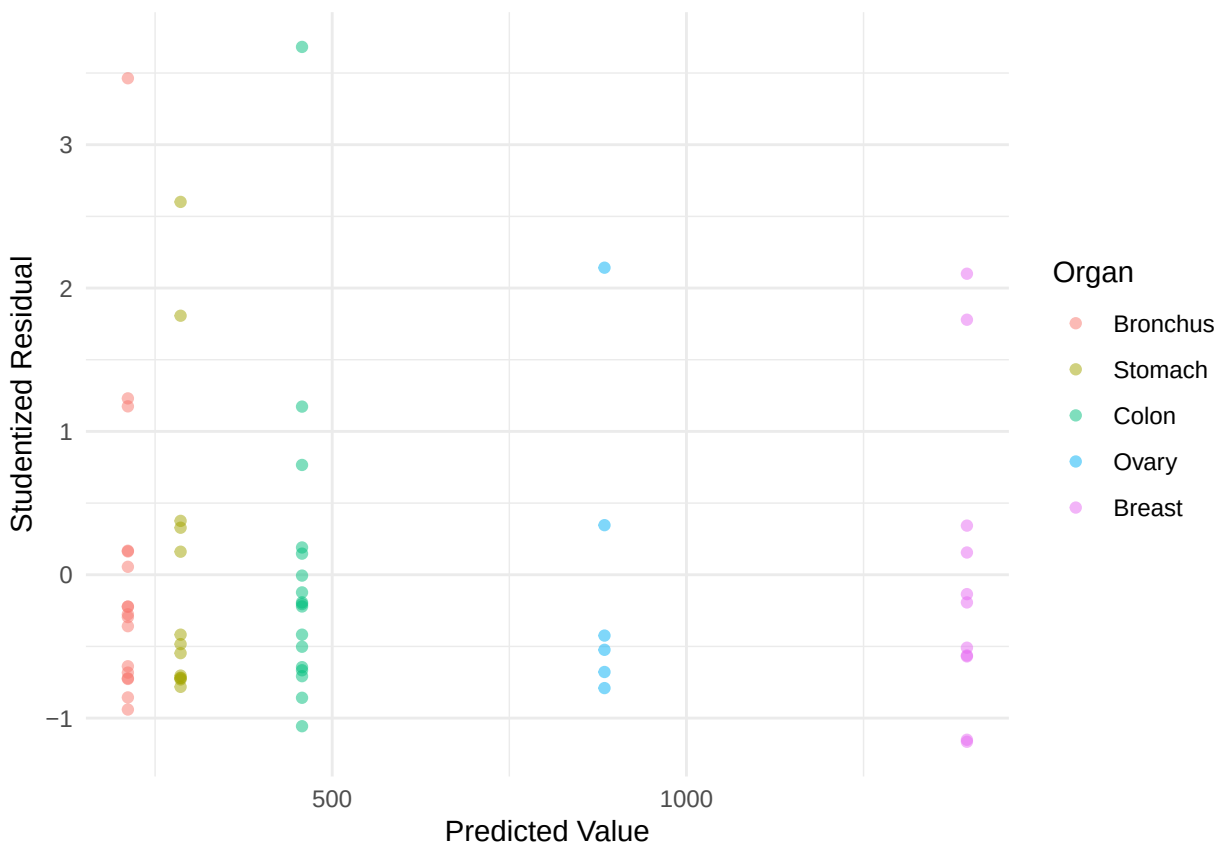
```

m.wls <- lm(Survival ~ Organ, weights = w, data = CancerSurvival)

CancerSurvival$yhat <- predict(m.wls)
CancerSurvival$rest <- rstudent(m.wls)

p <- ggplot(CancerSurvival, aes(x = yhat, y = rest, color = Organ)) +
  geom_point(alpha = 0.5) + theme_minimal() +
  labs(x = "Predicted Value", y = "Studentized Residual")
plot(p)

```



Note how this affects our inferences.

```

cbind(summary(m.ols)$coefficients, confint(m.ols))

```

	Estimate	Std. Error	t value	Pr(> t)	2.5 %	97.5 %
(Intercept)	211.59	162.4	1.3030	0.1976373	-113.34	536.5
OrganStomach	74.41	246.7	0.3017	0.7639784	-419.20	568.0
OrganColon	245.82	229.6	1.0704	0.2887820	-213.70	705.3
OrganOvary	672.75	317.9	2.1160	0.0385749	36.56	1308.9
OrganBreast	1184.32	259.1	4.5713	0.0000253	665.91	1702.7

```
cbind(summary(m.wls)$coefficients, confint(m.wls))
```

	Estimate	Std. Error	t value	Pr(> t)	2.5 %	97.5 %
(Intercept)	211.59	50.9	4.1571	0.0001057	109.74	313.4
OrganStomach	74.41	108.7	0.6846	0.4963078	-143.10	291.9
OrganColon	245.82	115.4	2.1296	0.0373858	14.85	476.8
OrganOvary	672.75	451.4	1.4904	0.1414343	-230.45	1575.9
OrganBreast	1184.32	377.0	3.1413	0.0026291	429.92	1938.7

```
organs <- unique(CancerSurvival$Organ)
trtools::contrast(m.ols, a = list(Organ = organs), cnames = organs)
```

	estimate	se	lower	upper	tvalue	df	pvalue
Stomach	286.0	185.7	-85.57	657.6	1.540	59	1.289e-01
Bronchus	211.6	162.4	-113.34	536.5	1.303	59	1.976e-01
Colon	457.4	162.4	132.48	782.3	2.817	59	6.587e-03
Ovary	884.3	273.3	337.39	1431.3	3.235	59	1.993e-03
Breast	1395.9	201.9	991.96	1799.9	6.915	59	3.770e-09

```
trtools::contrast(m.wls, a = list(Organ = organs), cnames = organs)
```

	estimate	se	lower	upper	tvalue	df	pvalue
Stomach	286.0	96.05	93.81	478.2	2.978	59	0.0042091
Bronchus	211.6	50.90	109.74	313.4	4.157	59	0.0001057
Colon	457.4	103.60	250.10	664.7	4.415	59	0.0000437
Ovary	884.3	448.49	-13.10	1781.8	1.972	59	0.0533281
Breast	1395.9	373.56	648.41	2143.4	3.737	59	0.0004228

```
trtools::contrast(m.ols,
  a = list(Organ = "Breast"),
  b = list(Organ = c("Bronchus", "Stomach", "Colon", "Ovary")),
  cnames = c("Breast vs Bronchus", "Breast vs Stomach",
    "Breast vs Colon", "Breast vs Ovary"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
Breast vs Bronchus	1184.3	259.1	665.9	1703	4.571	59	0.0000253
Breast vs Stomach	1109.9	274.3	561.1	1659	4.046	59	0.0001533
Breast vs Colon	938.5	259.1	420.1	1457	3.622	59	0.0006083
Breast vs Ovary	511.6	339.8	-168.4	1192	1.506	59	0.1375263

```
trtools::contrast(m.wls,
  a = list(Organ = "Breast"),
  b = list(Organ = c("Bronchus", "Stomach", "Colon", "Ovary")),
  cnames = c("Breast vs Bronchus", "Breast vs Stomach",
    "Breast vs Colon", "Breast vs Ovary"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
Breast vs Bronchus	1184.3	377.0	429.9	1939	3.1413	59	0.002629
Breast vs Stomach	1109.9	385.7	338.1	1882	2.8776	59	0.005572
Breast vs Colon	938.5	387.7	162.8	1714	2.4209	59	0.018577
Breast vs Ovary	511.6	583.7	-656.4	1680	0.8765	59	0.384340

Here's how you can do the comparison of one level with all others using the `contrast` function from the `emmeans` package.

```
library(emmeans)
emmeans::contrast(emmeans(m.wls, ~ Organ), method = "trt.vs.ctrl", ref = "Breast",
```

```
reverse = TRUE, adjust = "none", infer = TRUE)
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Breast - Bronchus	1184	377	59	430	1939	3.141	0.0026
Breast - Stomach	1110	386	59	338	1882	2.878	0.0056
Breast - Colon	938	388	59	163	1714	2.421	0.0186
Breast - Ovary	512	584	59	-656	1680	0.876	0.3843

Confidence level used: 0.95

Another approach is to assume that the variance of the response variable is some function of its expected response, and thus the weights are a function of the expected response. With right-skewed response variables one common functional relationship is that

$$\text{Var}(Y_i) \propto E(Y_i),$$

or, more generally,

$$\text{Var}(Y_i) \propto E(Y_i)^p,$$

where p is some power (usually $p \geq 1$). So the weights would then be

$$w_i \propto \frac{1}{E(Y_i)^p}.$$

We do not know $E(Y_i)$, but $\hat{y}_i$ is an estimate of $E(Y_i)$. But we need the weights to compute $\hat{y}_i$!

Two situations:

1. Estimates of the model parameters and thus $\hat{y}_i$ *do not* depend on the weights.¹ Here we can compute the weights with an initial regression model without weights.
2. Estimates of the model parameters and thus $\hat{y}_i$ *do* depend on the weights. An approach we can use here is *iteratively weighted least squares*.

It can be shown that $\hat{y}_i$ does not depend on the weights for the model for the `CancerSurvival` model. We can use the estimates from ordinary least squares to obtain weights of $w_i = 1/\hat{y}_i^p$.

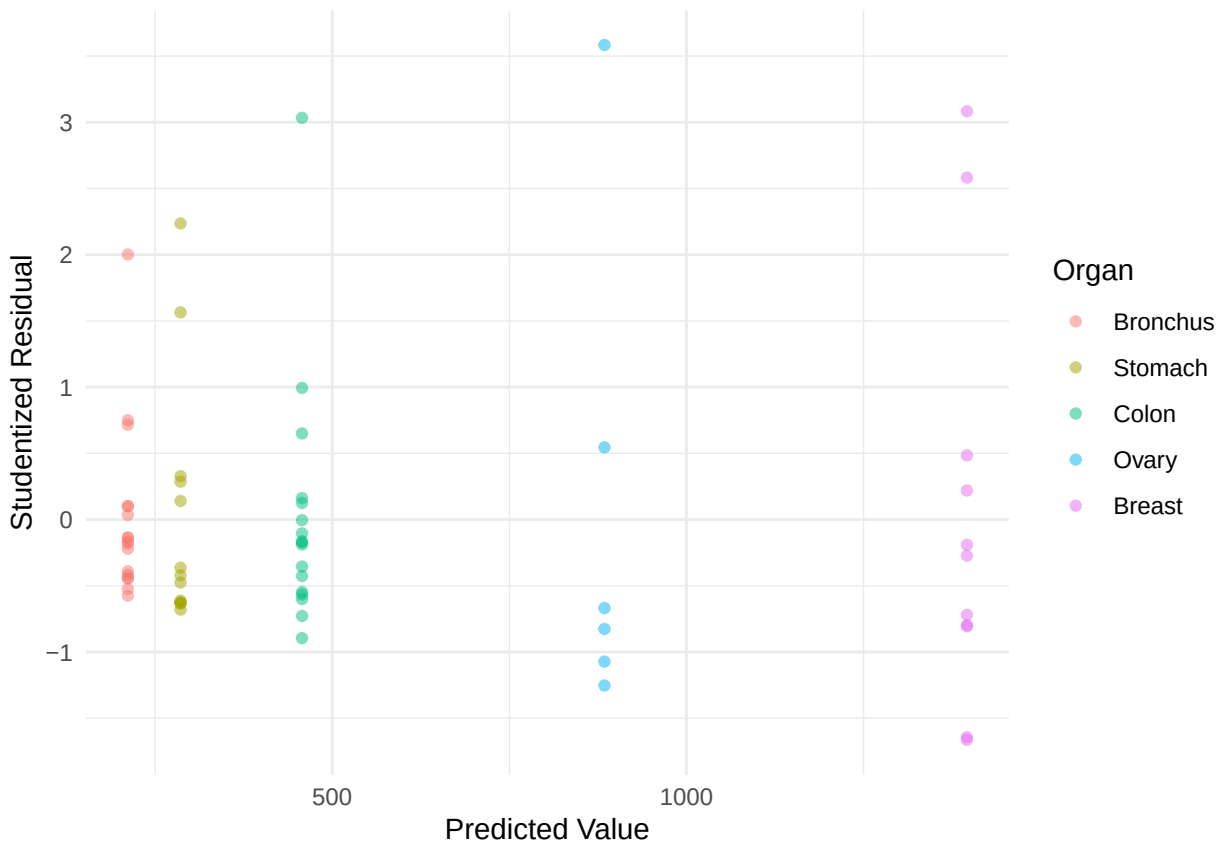
```
m.ols <- lm(Survival ~ Organ, data = CancerSurvival)

CancerSurvival$w <- 1/predict(m.ols)
m.wls <- lm(Survival ~ Organ, data = CancerSurvival, weights = w)

CancerSurvival$yhat <- predict(m.wls)
CancerSurvival$rest <- rstudent(m.wls)

p <- ggplot(CancerSurvival, aes(x = yhat, y = rest, color = Organ)) +
  geom_point(alpha = 0.5) + theme_minimal() +
  labs(x = "Predicted Value", y = "Studentized Residual")
plot(p)
```

¹In practice the most common situation where this will happen is with models with only a single categorical explanatory variable (or models with multiple categorical explanatory variables that include interactions).



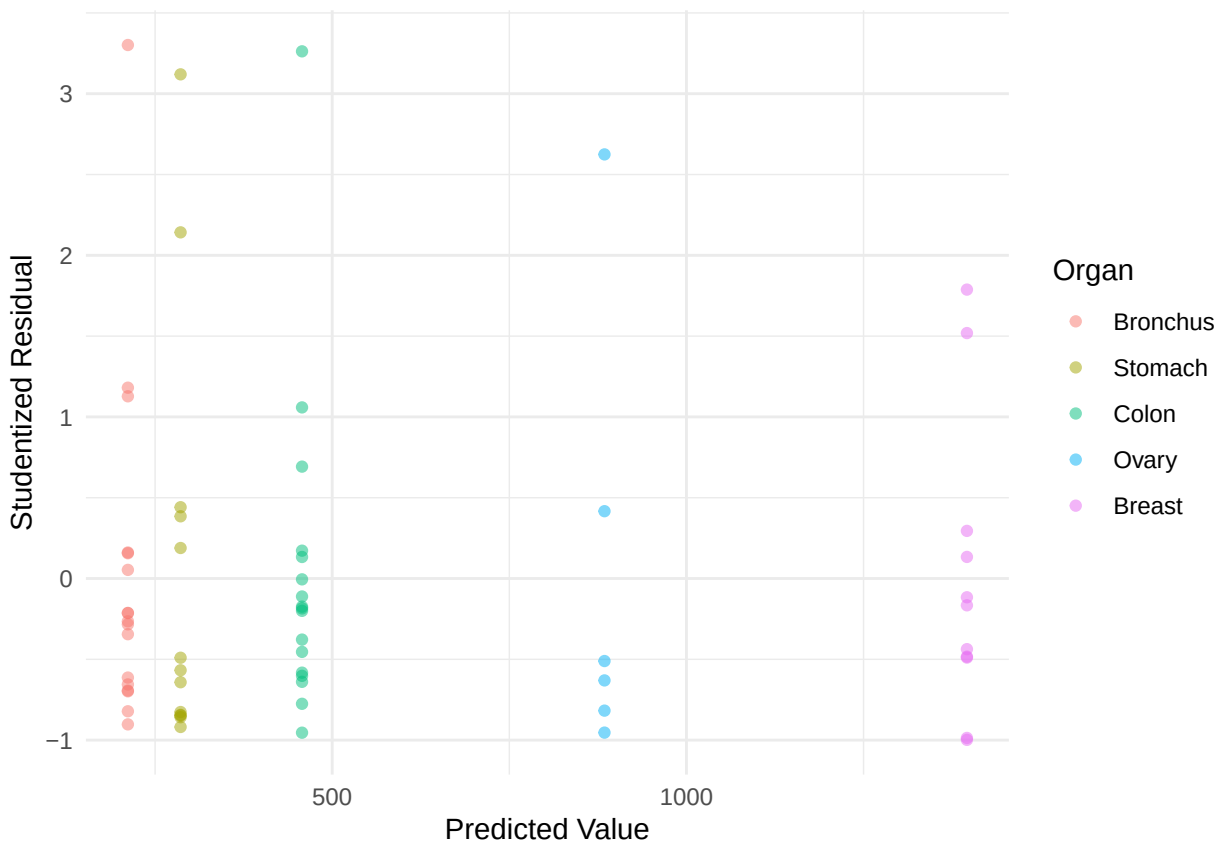
Maybe we could do better. Let's try $p = 2$ — i.e., $\text{Var}(Y_i) \propto E(Y_i)^2$.

```
m.ols <- lm(Survival ~ Organ, data = CancerSurvival)

CancerSurvival$w <- 1/predict(m.ols)^2
m.wls <- lm(Survival ~ Organ, data = CancerSurvival, weights = w)

CancerSurvival$yhat <- predict(m.wls)
CancerSurvival$rest <- rstudent(m.wls)

p <- ggplot(CancerSurvival, aes(x = yhat, y = rest, color = Organ)) +
  geom_point(alpha = 0.5) + theme_minimal() +
  labs(x = "Predicted Value", y = "Studentized Residual")
plot(p)
```



Example: Consider again following data from a study on the effects of fuel reduction on biomass.

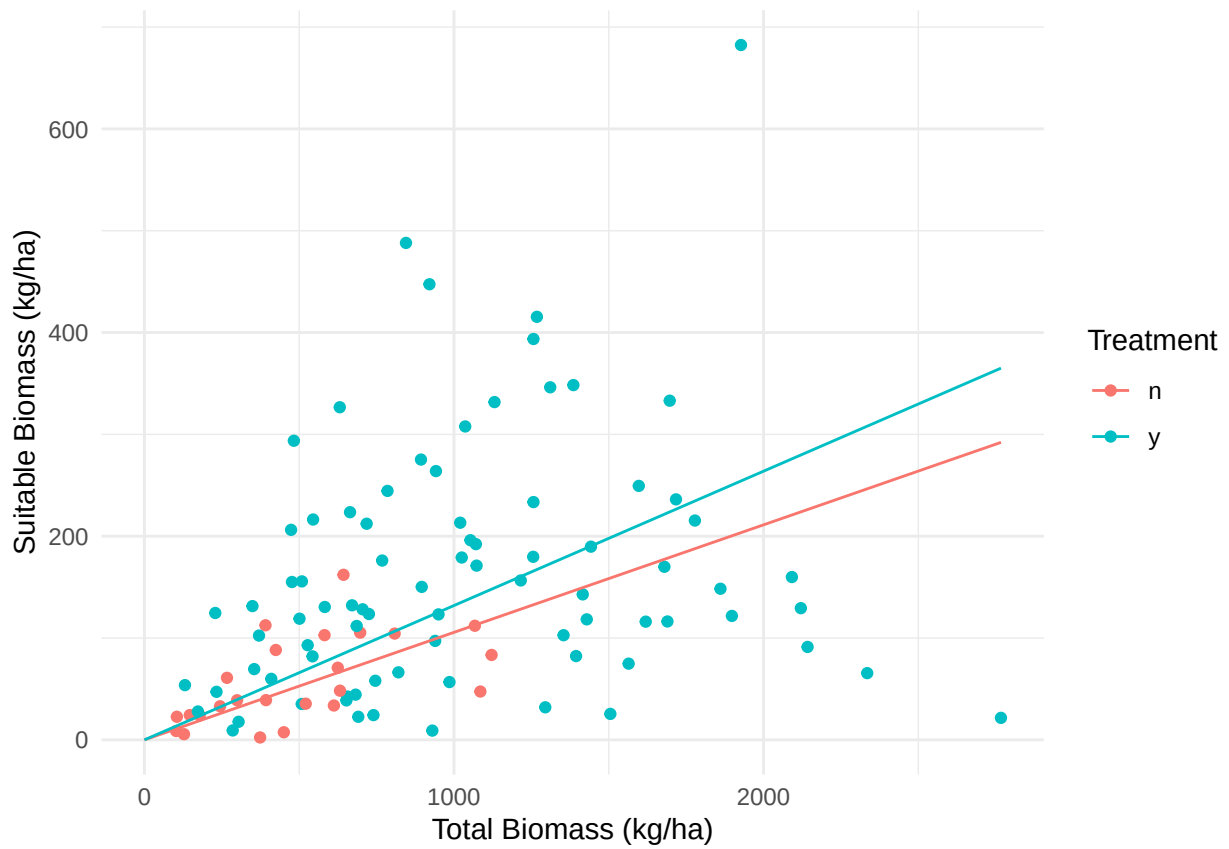
```
library(trtools) # for biomass data
```

```
m <- lm(suitable ~ -1 + treatment:total, data = biomass)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
treatmentn:total	0.1056	0.04183	2.524	1.31e-02
treatmenty:total	0.1319	0.01121	11.773	7.61e-21

```
d <- expand.grid(treatment = c("n","y"), total = seq(0, 2767, length = 10))
d$yhat <- predict(m, newdata = d)
```

```
p <- ggplot(biomass, aes(x = total, y = suitable, color = treatment)) +
  geom_point() + geom_line(aes(y = yhat), data = d) + theme_minimal() +
  labs(x = "Total Biomass (kg/ha)",
       y = "Suitable Biomass (kg/ha)",
       color = "Treatment")
plot(p)
```



```
biomass$yhat <- predict(m)
biomass$rest <- rstudent(m)

p <- ggplot(biomass, aes(x = yhat, y = rest, color = treatment)) +
  geom_point() + theme_minimal() +
  labs(x = "Predicted Value",
       y = "Studentized Residual",
       color = "Treatment")
plot(p)
```



Here we might also assume that $\text{Var}(Y_i) \propto E(Y_i)^p$, with weights of $w_i = 1/\hat{y}_i$. But here things are a bit more complicated for this model: the w_i depend on the $\hat{y}_i$, the $\hat{y}_i$ depend on the w_i . In the model for the `CancerSurvival` data this was not an issue because there the estimates of the model parameters, and thus $\hat{y}_i$, did not depend on the weights so we could use ordinary least squares where all $w_i = 1$ to get the $\hat{y}_i$. But that is not true for this model. But we can solve this problem using *iteratively weighted least squares*.

```
biomass$w <- 1 # initial weights are all equal to one
for (i in 1:5) {
  m.wls <- lm(suitable ~ -1 + treatment:total, weights = w, data = biomass)
  print(coef(m.wls)) # optional
  biomass$w <- 1 / predict(m.wls)
}
```

```
treatmentn:total treatmenty:total
      0.1056      0.1319
treatmentn:total treatmenty:total
      0.1155      0.1578
treatmentn:total treatmenty:total
      0.1155      0.1578
treatmentn:total treatmenty:total
      0.1155      0.1578
treatmentn:total treatmenty:total
      0.1155      0.1578
```

```
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
treatmentn:total	0.1056	0.04183	2.524	1.31e-02
treatmenty:total	0.1319	0.01121	11.773	7.61e-21


```
summary(m.wls)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
treatmentn:total	0.1155	0.02887	4.00	1.189e-04
treatmenty:total	0.1578	0.01268	12.45	2.501e-22